

# Audio Circuit Engineering Calculations

## Impedance Bridging

$$Z_{\text{load}} \geq 10 \times Z_{\text{source}}$$

## Thermal Noise Voltage

$$v_n = \sqrt{4k_B T R \Delta f}$$

## 833:1 Divider

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_{\text{shunt}}}{R_{\text{series}} + R_{\text{shunt}}}$$

## Preamp Input Transformer Ratio

$$Z_{\text{primary}} = Z_{\text{secondary}} \cdot \frac{25}{37500} = 1500 \Omega$$

The Opamp provides a secondary load resistance of  $37.5 \Omega$ , which is the input impedance of the preamp. The primary load resistance is  $1500 \Omega$ , which is the output impedance of the opamp. The transformer ratio is 25:1, which means that the voltage at the primary side is 25 times the voltage at the secondary side.

## Impedance Ratio Formula

$$\frac{Z_{\text{primary}}}{Z_{\text{secondary}}} = \left( \frac{N_p}{N_s} \right)^2$$

## Line Output Ratio

$$(N_p : N_s): 2:1$$

$$\text{Voltage Loss (dB)} = 20 \cdot \log_{10}(0.5) \approx -6.02 \text{ dB}$$

## Op-amp Stability Capacitors

$$f_{\text{filter}} = \frac{1}{2\pi \cdot R_f \cdot C_f} = \frac{1}{2\pi \cdot (20000 \Omega) \cdot (22 \times 10^{-12} \text{ F})} \approx 361.7 \text{ kHz}$$

The feedback resistor is set to 20k $\Omega$ .

## 17 Channel Panning DAC-to-VCA Step-Down Divider

$$V_{\text{out}} = 5\text{V} \cdot \left( \frac{12 \Omega}{10000 \Omega + 12 \Omega} \right) = 5\text{V} \cdot \left( \frac{12}{10012} \right) \approx 5.99 \text{ mV}$$

The THAT 4305 engine alters gain a minute long scale gate metric of exactly  $-6.1 \text{ mV}$  per decibel (dB).

## Noise Gate / Expander Circuit Calculations

### Knob at Minimum (0 $\Omega$ Pot + 10k $\Omega$ Resistor)

$$\tau = 10000 \Omega \cdot (1 \times 10^{-6} \text{ F}) = 0.01 \text{ seconds}$$

$$\text{Minimum Release} = 5 \cdot 0.01\text{s} = \mathbf{50} \text{ milliseconds}$$

### Knob at Maximum (50k $\Omega$ Pot + 10k $\Omega$ Resistor)

$$\tau = 60000 \Omega \cdot (1 \times 10^{-6} \text{ F}) = 0.06 \text{ seconds}$$

$$\text{Maximum Release} = 5 \cdot 0.06\text{s} = \mathbf{300} \text{ milliseconds}$$

## Calculating Frequency Sweeps

### Knob at max resistance (10k $\Omega$ Pot + 1.2k $\Omega$ Resistor)

$$f_c = \frac{1}{2\pi \cdot (11200 \Omega) \cdot (220 \times 10^{-9} \text{ F})} \approx 64.5 \text{ Hz}$$

### Knob at min resistance (0 $\Omega$ Pot + 1.2k $\Omega$ Resistor)

$$f_c = \frac{1}{2\pi \cdot (1200 \Omega) \cdot (220 \times 10^{-9} \text{ F})} \approx 602.8 \text{ kHz}$$